

DocMind AI — Complete Architecture, Workflow & Technical Guide

DocMind AI — Complete Architecture, Workflow & Technical Guide

1. Executive Summary & Product Vision

DocMind AI is an enterprise-grade, multimodal, trilingual document intelligence and studio platform. It empowers users to converse with, analyze, summarize, compare, augment, and listen to complex documents across multiple formats without hallucination.

Key Objectives

- Universal Ingestion: Ingest PDF, Excel (.xlsx), PowerPoint (.pptx), Word (.docx), CSV, and Plain Text.
- Trilingual Reasoning: Complete fluency and native script grounding in English, Hindi (■■■■■■■), and Gujarati (■■■■■■■).
- Strict Zero-Hallucination RAG: Reciprocal Rank Fusion (RRF) combining dense semantic vectors with BM25 sparse keyword search.
- Proactive Compliance Alerts: Automated scanning of notices and contracts for required actions, urgency levels, deadlines, and penalty consequences.
- Document Studio: Prompt-based modular augmentation allowing users to draft clauses, append appendices, and export directly to professional DOCX and PDF documents.
- Multi-User Credentials & Session Privacy: Salted PBKDF2 cryptographic authentication, isolated chat histories per user, and a 1-click demo account.

2. End-to-End System Architecture & Workflow

Processing Workflow Pipeline

1. Document Ingestion Layer:
 - The user uploads a file through the responsive Streamlit interface or the FastAPI /upload endpoint.
 - Universal Loaders extract structured text, Markdown tables, presentation slide hierarchies, or spreadsheets.
2. Chunking & Multilingual Vectorization:
 - Chunking engine segments documents into overlapping 1000-character passages with metadata preservation.
 - Gemini Embedding Model (gemini-embedding-001) produces 3072-dimensional dense vectors in batch.
3. Dual-Engine Storage & Indexing:
 - ChromaDB stores vector embeddings for dense semantic similarity search.
 - BM25Okapi maintains a sparse lexical index for exact keywords, numbers, dates, and contract clause IDs.
4. Intelligent Query & RAG Engine:
 - Query Rewriter resolves conversational follow-ups and pronouns using past chat history.
 - RRF Hybrid Search scores and ranks candidate chunks.
 - Cascade Resilience Layer queries gemini-flash-lite-latest with system instructions for structured grounding and exact citations.

5. Post-Processing & Multimodal Delivery:

- Proactive Action & Deadline Extraction flags compliance risks.
- Text-to-Speech (gTTS) synthesizes answers into audio for voice listening.
- Document Studio exports augmented documents into DOCX or PDF.

3. Technology Stack, Libraries & AI Models

AI Models & Foundations

- Inference & Reasoning LLM: Google Gemini Flash Lite (gemini-flash-lite-latest). Ultra-fast latency (<2.5s), cost-effective token usage, and superior multilingual capability.
- Vector Embedding Model: Google Gemini Embedding (gemini-embedding-001). 3072 dimensions, native batch embedding API.
- Failover Cascade Engine: Automated retry mechanism with exponential backoff and model cascade failover to guarantee 99.9% uptime.

Core Libraries & Frameworks

- Backend API: FastAPI, Uvicorn, Pydantic, HTTPX.
- Frontend UI: Streamlit, Custom HTML/CSS Component bridge, BaseWeb Tabs.
- Vector Database & Retrieval: ChromaDB (Persistent client), Rank-BM25 (BM25Okapi).
- Document Loaders: PyPDF / PyMuPDF (PDF), openpyxl (Excel), python-pptx (PowerPoint), python-docx (Word).
- Export Engines: python-docx (DOCX builder), reportlab (PDF rendering with Platypus).
- Speech & Audio: gTTS (Google Text-to-Speech MP3 generator), edge-tts, pyttsx3 fallback.
- Security & Credentials: SQLite3 (users.db), hashlib (PBKDF2-HMAC-SHA256), hmac (constant-time digest).

4. Key Features Implemented

1. ChatGPT-Style Persistent Chat

- Multi-session chat history with title auto-generation based on the first prompt.
- User-scoped chat files (chat_sessions_{user_id}.json) preventing data leaks between accounts.
- Clear chat, new conversation button, and session deletion controls.

2. Proactive Action & Deadline Detection

- Analyzes incoming documents for mandatory obligations.
- Classifies documents into High, Medium, or Low urgency.
- Surfaces specific action items, due dates, and financial/legal consequences if missed.

3. Document Studio (Augment & Export)

- Modular drafting: Prompts draft only the requested additions or revisions, preserving token budget.
- Clean 1-click export to formatted Word (.docx) or Adobe PDF (.pdf) documents.

4. Text-to-Speech (Audio Voice Reader)

- Multilingual voice reader supporting English, Hindi, and Gujarati.
- High-efficiency in-memory audio player with dynamic audio rendering.

5. Multi-User Authentication & Security

- Secure registration and login with salt-based password protection.
- Pre-configured demo user (demo@docmind.ai / Demo@123).
- Clean sign-out and session reset.

6. Full Responsiveness

- Fully responsive styling tailored for Mobile Phones ($\leq 600\text{px}$), Tablets ($601\text{px} - 1024\text{px}$), and Desktops ($> 1024\text{px}$).

5. Future Roadmap & Enhancements (What to Build Next)

Proposed Advanced Features

1. Optical Character Recognition (OCR) for Handwritten & Scanned Documents:
 - Integrate Tesseract or PaddleOCR to read photographed paper receipts, handwritten signatures, and low-resolution scans.
2. Cloud Storage Sync Integrations:
 - Direct connectors for Google Drive, Microsoft OneDrive, Dropbox, and Amazon S3 buckets.
3. Multi-Agent Collaborative Review:
 - Specialized agents: Legal Risk Auditor, Financial Ratio Analyst, and Compliance Reviewer debating document validity.
4. Interactive Analytics & Data Visualizations:
 - Automatic generation of Plotly charts and interactive KPI dashboards directly from Excel and CSV documents.
5. Role-Based Access Control (RBAC):
 - Granular organizational tiers: Administrator, Financial Auditor, Department Editor, and Read-Only Guest.
6. Real-Time Token Streaming (WebSockets):
 - Live character-by-character typewriter streaming for instant user feedback.

6. How to Deploy & Maintain

- Local Development: Run FastAPI on port 8000 and Streamlit on port 8501.
- Containerization: Docker multi-stage build running start.sh with supervisor.
- Hugging Face Spaces / Cloud VM: Deployable with zero modifications using Docker SDK.